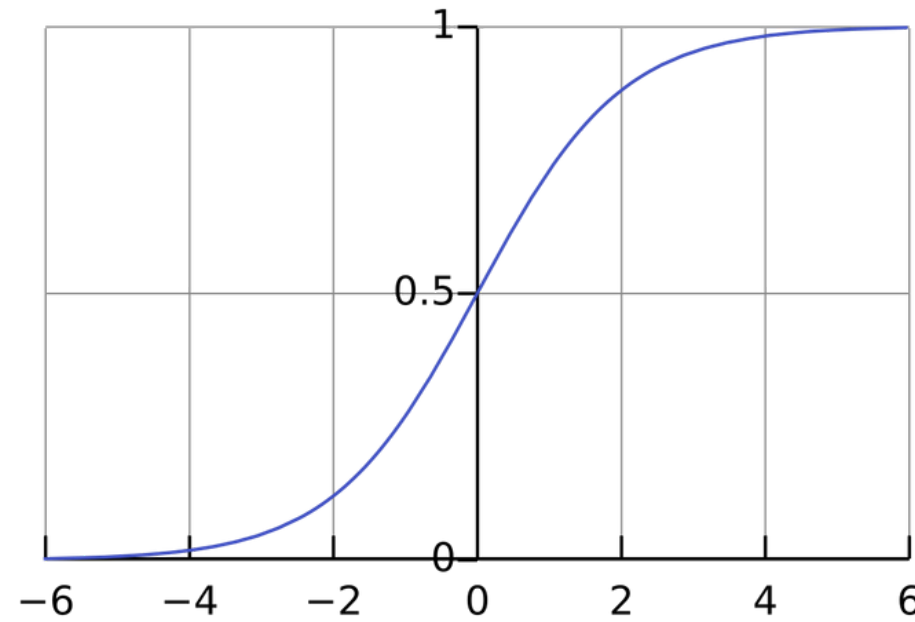




Activation Function

2. Sigmoid

$$f(z) = \frac{1}{1 + e^{-z}}$$



Intuition

- Smoothly squashes values between 0 and 1.

Pros

- Probabilistic interpretation
- Used in binary classification

Cons

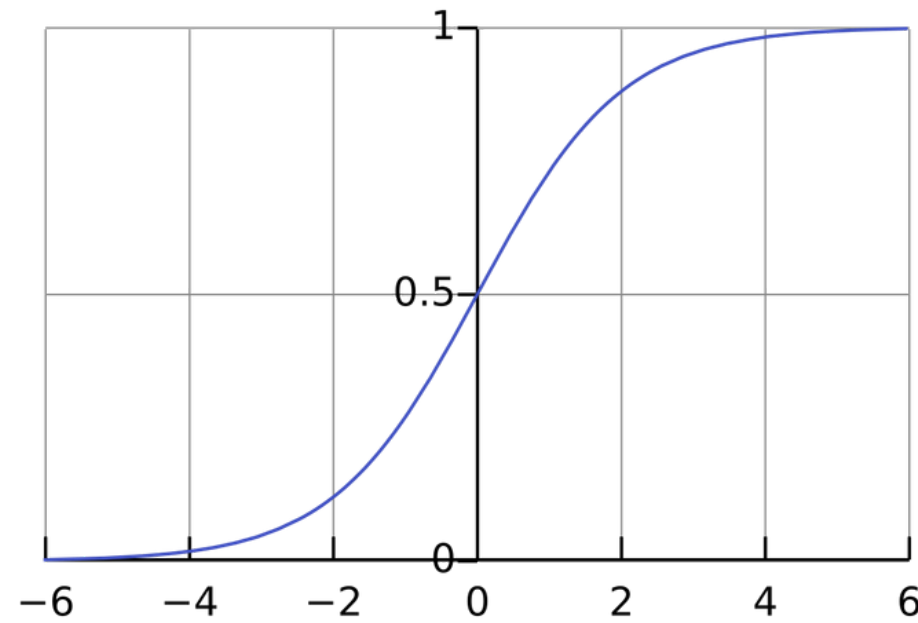
- Vanishing gradients
- Not zero-centered



Activation Function

2. Sigmoid

$$f(z) = \frac{1}{1 + e^{-z}}$$



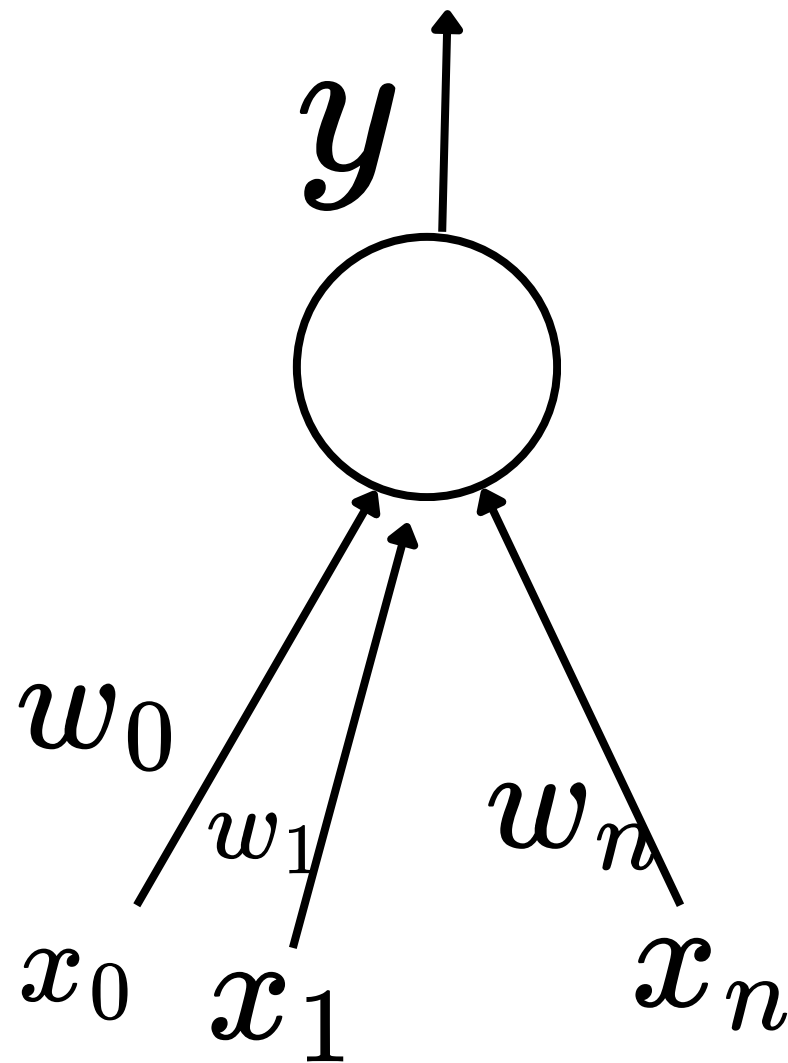
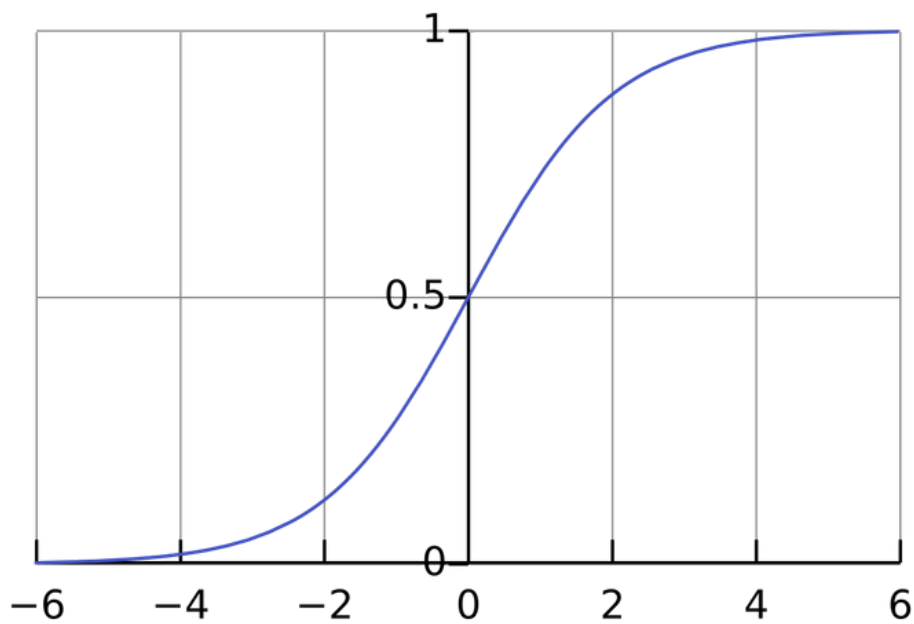
- The problem with threshold logic is, it is very harsh, 0 or 1
- But, for many cases we need soft answer



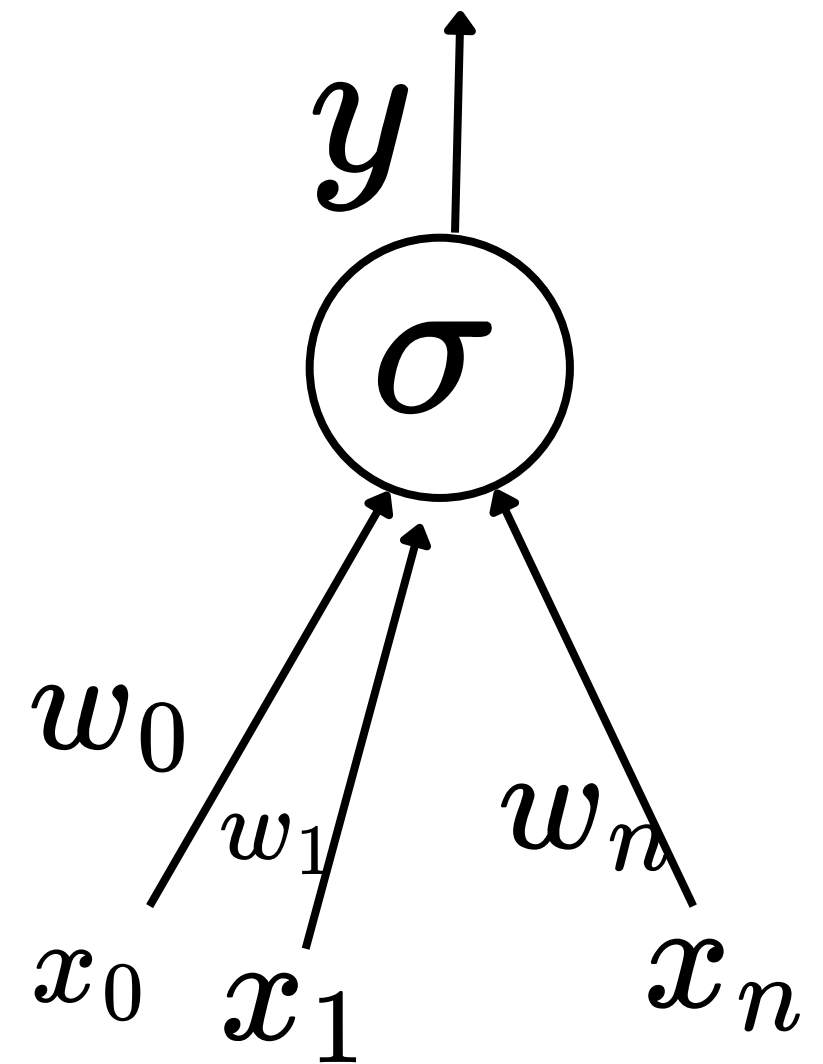
Activation Function

2. Sigmoid

$$f(z) = \frac{1}{1 + e^{-z}}$$



$$y = 0 \quad \sum_{i=0}^n w_i x_i \geq 0$$
$$y = 1 \quad \sum_{i=0}^n w_i x_i < 0$$



$$y = \frac{1}{1 + e^{-\sum_{i=0}^n w_i x_i}}$$